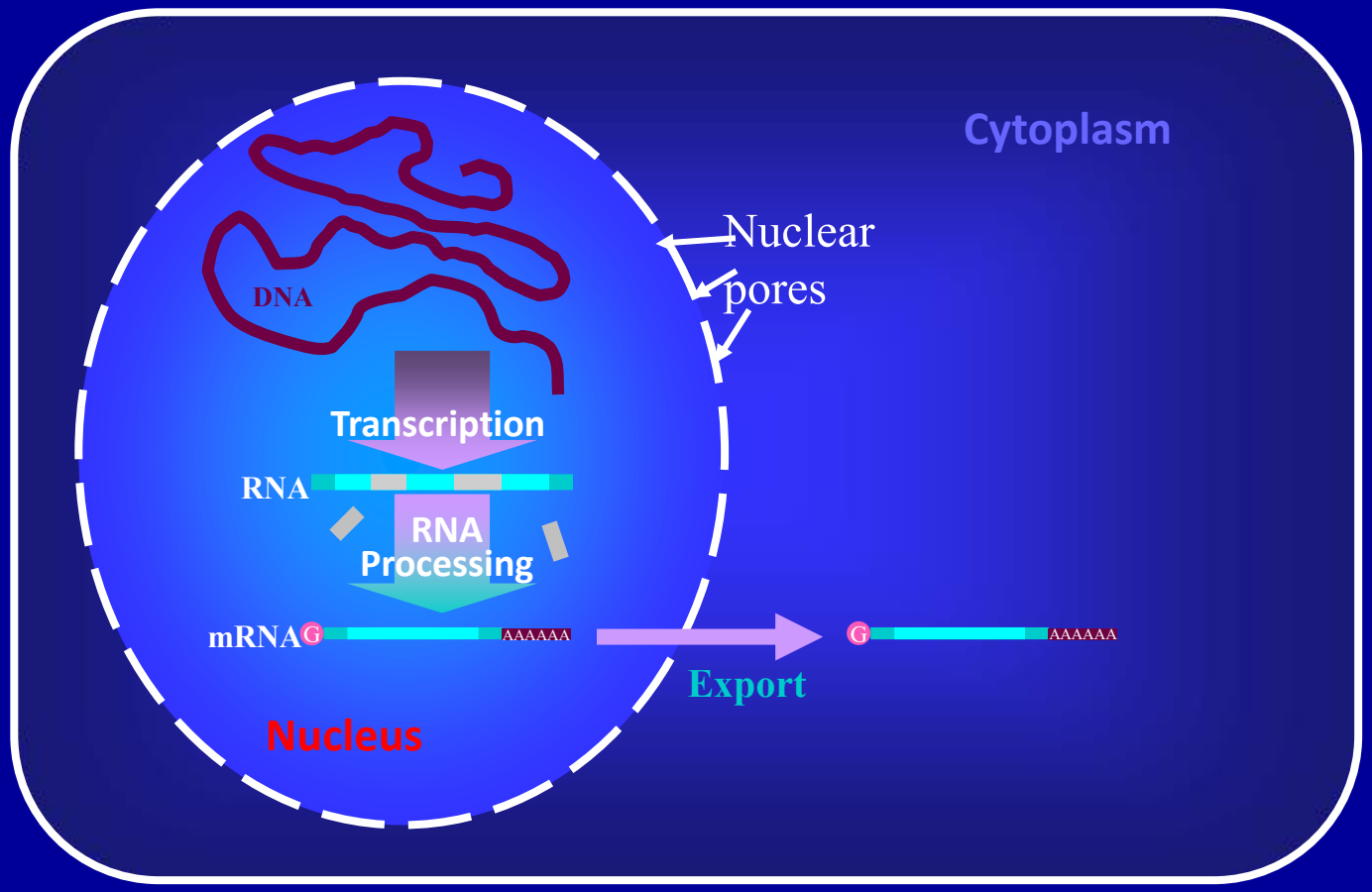
Eukaryotic mRNA Transcripts are Processed



One Gene, One Polypeptide Hypothesis

- In general, one gene generates one polypeptide:
  - Specific mutations in the bread mold *Neurospora* resulted in the loss of a specific enzyme in a biochemical pathway
  - Certain human hereditary diseases are caused by the absence of a certain enzymes

First letter	Second letter: U	Second letter: C	Second letter: A	Second letter: G	Third letter
U	UUU, UUC: Phenyl-alanine	UCU, UCC, UCA, UCG: Serine	UAU, UAC: Tyrosine	UGU, UGC: Cysteine	U
	UUA, UUG: Leucine		UAA: Stop codon; UAG: Stop codon	UGA: Stop codon; UGG: Tryptophan	C, A, G
C	CUU, CUC, CUA, CUG: Leucine	CCU, CCC, CCA, CCG: Proline	CAU, CAC: Histidine	CGU, CGC, CGA, CGG: Arginine	U
			CAA, CAG: Glutamine		C, A, G
A	AUU, AUC, AUA: Isoleucine	ACU, ACC, ACA, ACG: Threonine	AAU, AAC: Asparagine	AGU, AGC: Serine	U
	AUG: Methionine; start codon		AAA, AAG: Lysine	AGA, AGG: Arginine	C, A, G
G	GUU, GUC, GUA, GUG: Valine	GCU, GCC, GCA, GCG: Alanine	GAU, GAC: Aspartic acid	GGU, GGC, GGA, GGG: Glycine	U
			GAA, GAG: Glutamic acid		C, A, G

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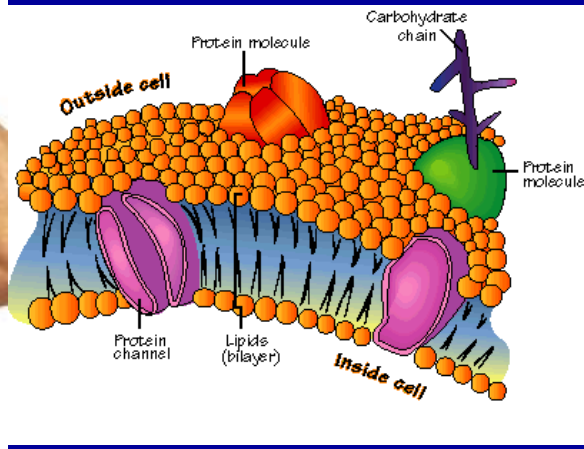
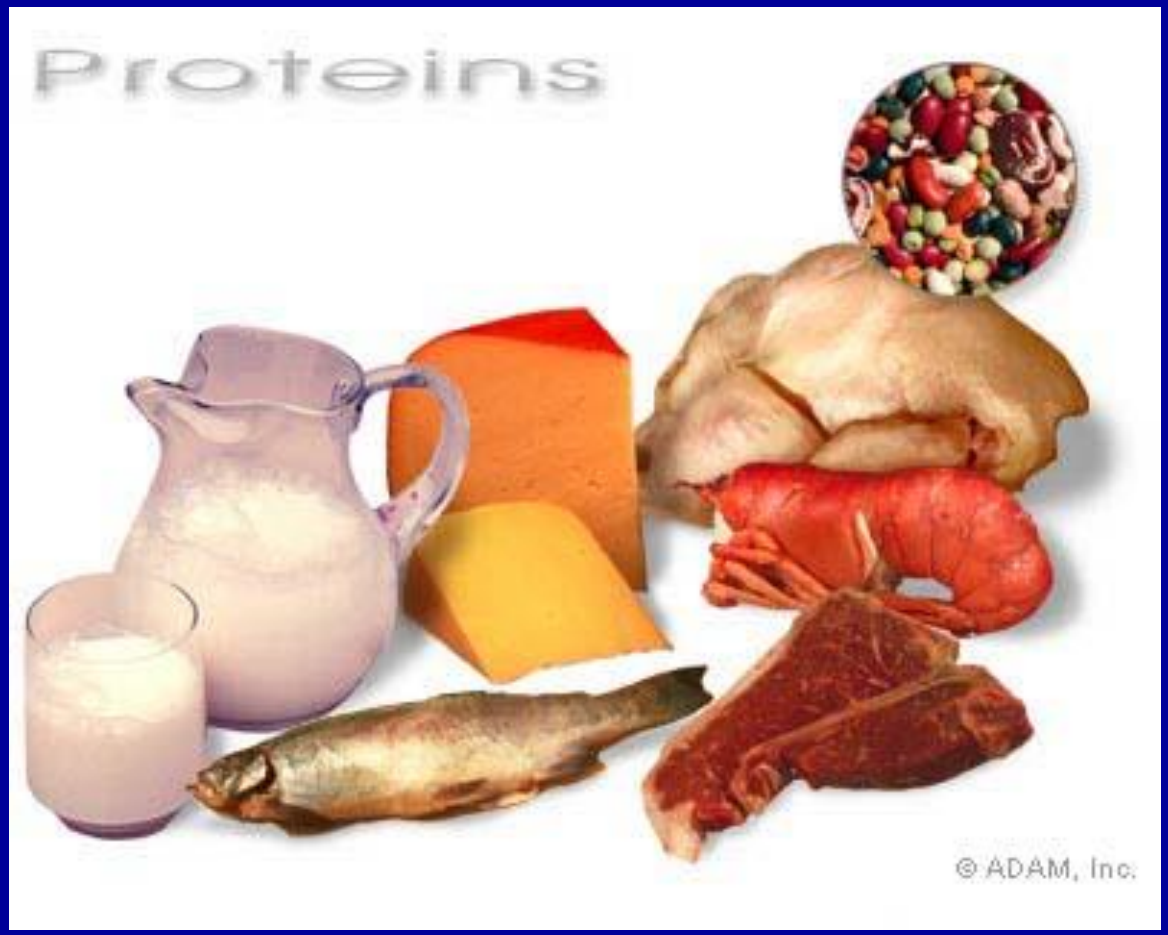
Figure 12.5

- Function of proteins:
  1. Help fight **disease**
  2. Build new body **tissue**
  3. **Enzymes** used for digestion and other chemical reactions are proteins

     (Enzymes **speed up** the **rate** of a reaction)
  4. Component of all **cell membranes**



These amino acids come from the **food we eat**. Proteins we eat are broken down into individual **amino acids** and then simply **rearranged** into new **proteins** according to the needs and directions of our **DNA**.



The Genetic Code

- The genetic code consists of triplets of nucleotides (codons)
- Since there are four bases (A,C,G,T), there are 64 possible codons (4^3):
  - 1 start codon (encodes methionine)
  - 3 termination codons (stop translation)
  - 60 codons for the 20 amino acids
- Hence, the genetic code is redundant; i.e. there is more than one codon for certain amino acids.
- However, a single codon does not specify more than one amino acid
- The genetic code is universal (almost)

The Central Dogma

- The Flow of Information: DNA → RNA → protein

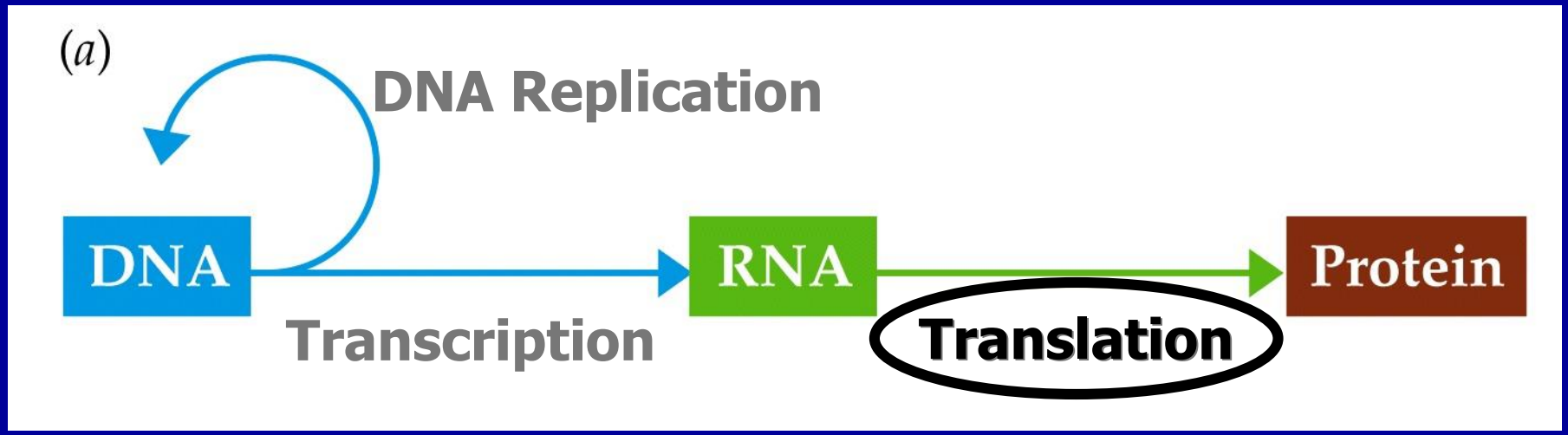


Figure 12.2

- Translation if the process whereby mRNA is decoded into protein.

Translation: mRNA directed protein synthesis

- During translation, the order in which amino acids are linked together is specified by the codons in the mRNA
- Transfer RNA (tRNA) acts as the adapter.
- Via the anticodon-codon interaction, tRNA brings the correct amino acid corresponding to the mRNA codon.
- Aminoacyl-tRNA synthetases, a family of activating enzymes, attach specific amino acids to their appropriate tRNA's, forming charged tRNA's.

The ribosome

- The mRNA meets the charged tRNA's at the ribosome
- The ribosome has two subunits: small and large
- The ribosome has four sites: E, P, A, T

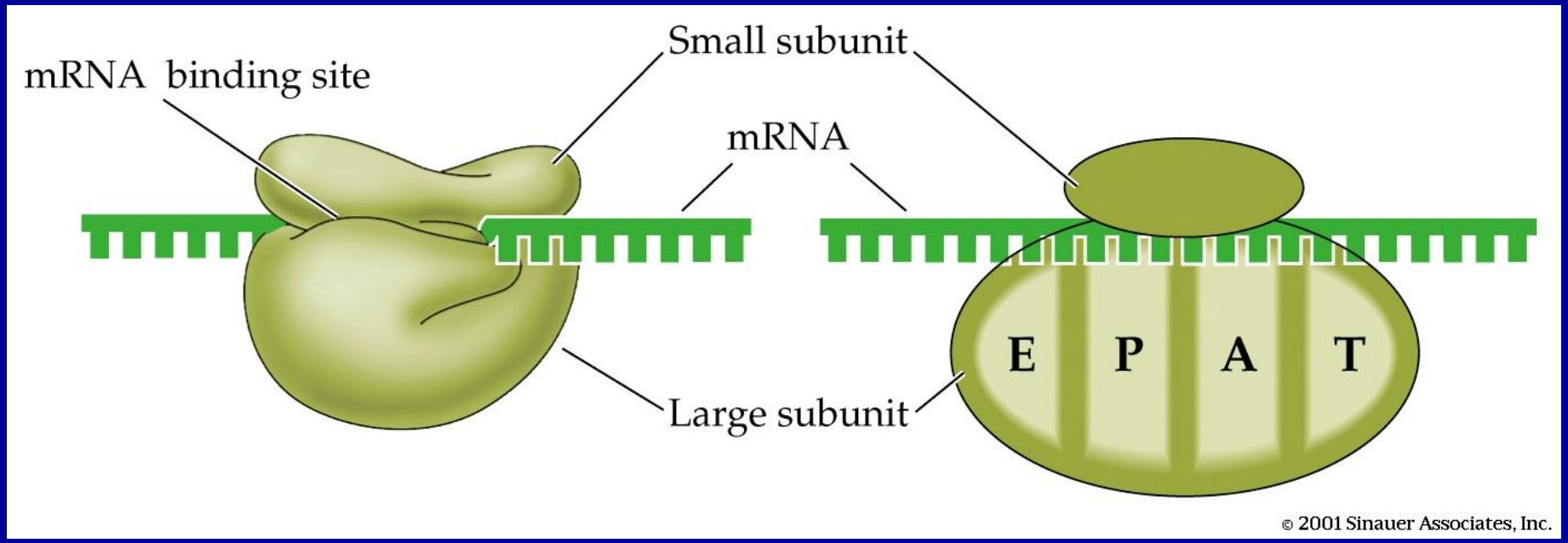
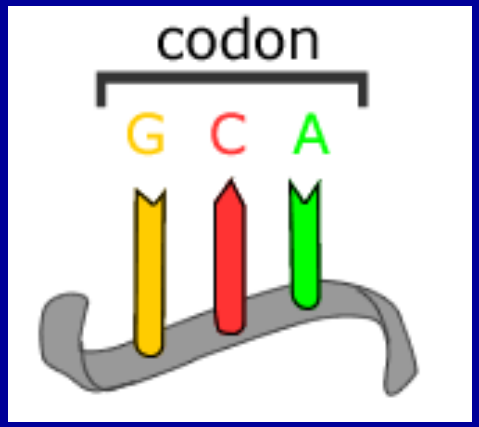


Figure 12.9

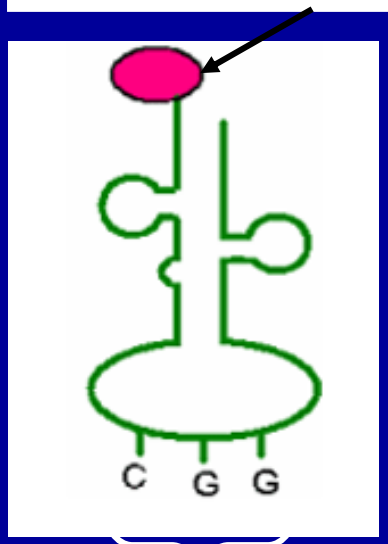
Translation - Initiation

- Translation has three phases:
  - Initiation
  - Elongation
  - Termination
- Initiation: An initiation complex forms, consisting of the initiator tRNA charged with methionine and the small ribosomal subunit bound to mRNA triggers the beginning of translation

- A series of **three** adjacent **bases** in an mRNA molecule codes for a specific amino acid—called a **codon**.



- Each **tRNA** has 3 nucleotides that are **complementary** to the **codon** in mRNA.



- Each **tRNA** codes for a **different** amino acid.

Translation initiation

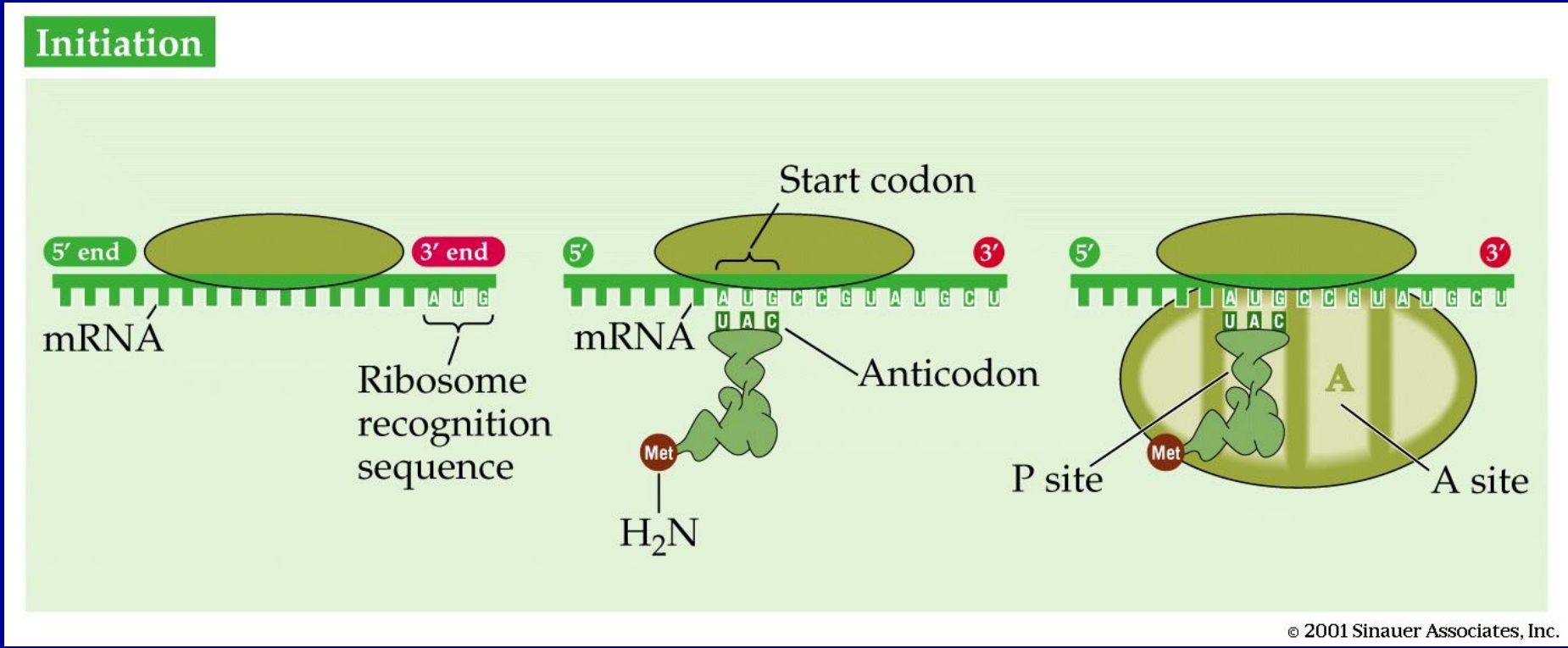


Figure 12.10

Translation - Elongation

- The ribosome moves along the mRNA one codon at a time in a 5′-to-3′ direction
- Polypeptides grow from the N terminus toward the C terminus
- Charged tRNA's bring amino acids to the ribosome sequentially
- Specificity is provided by:
  - the anticodon (tRNA) -codon (mRNA) interaction
  - the accuracy of aminoacyl-tRNA synthetases

Translation - Elongation

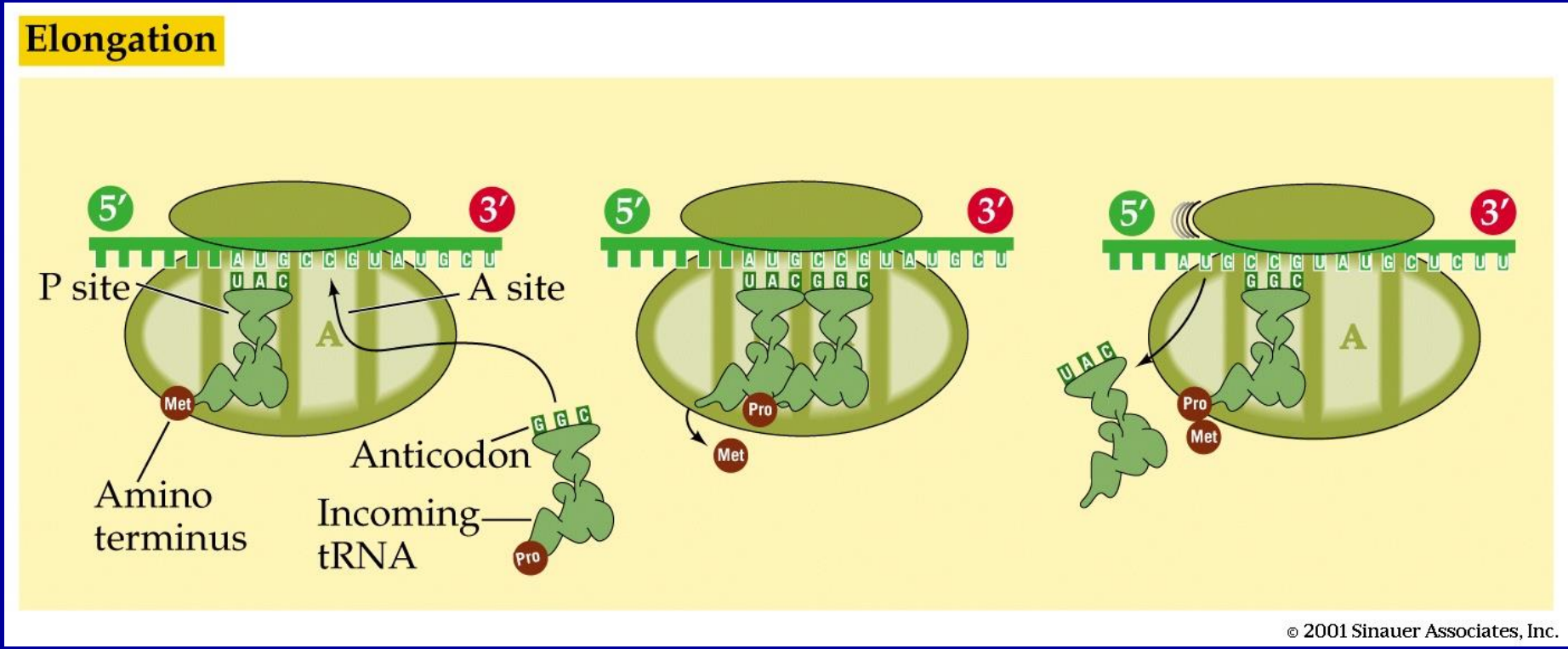


Figure 12.11 – Part 1

Translation - Elongation

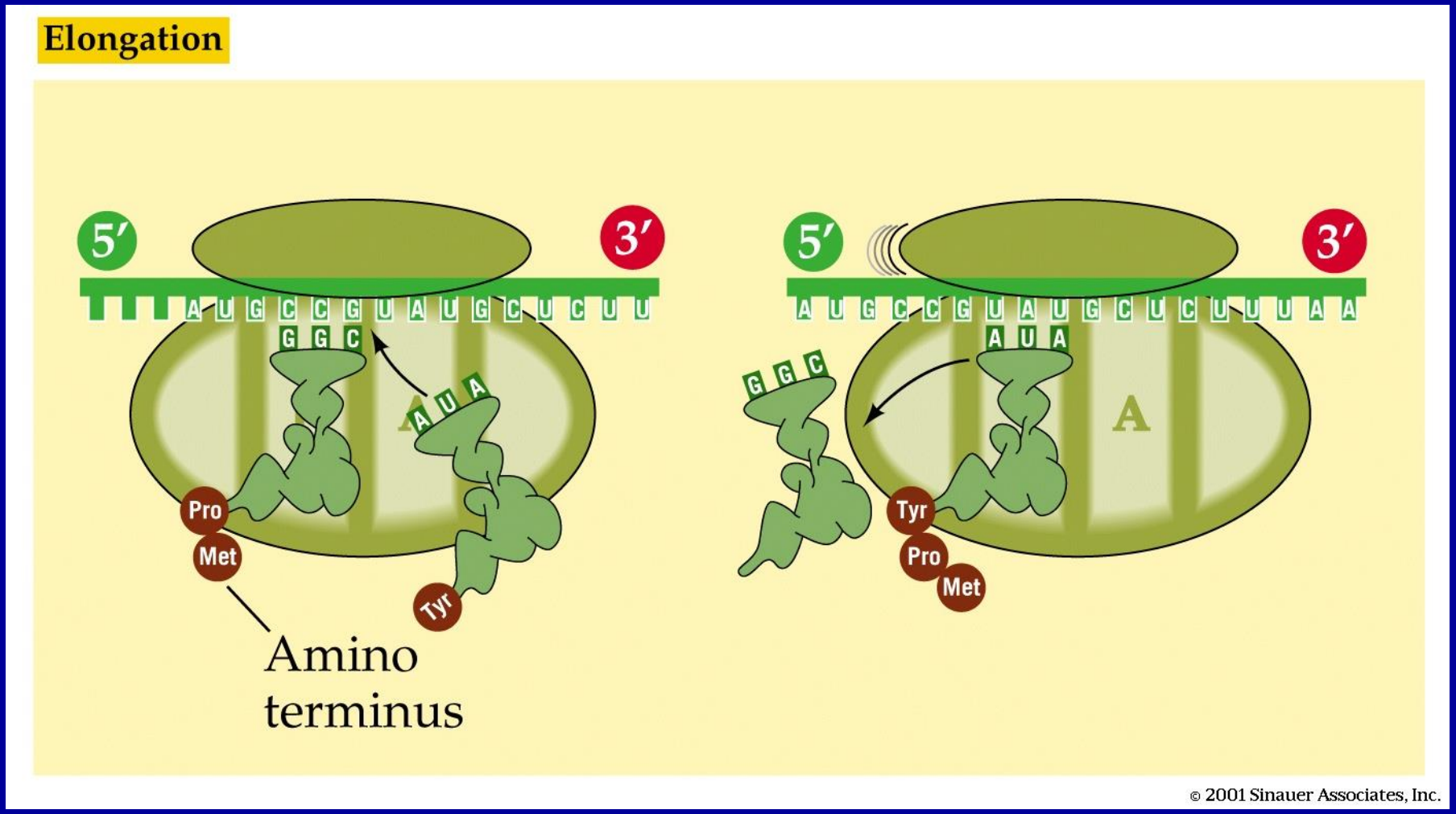


Figure 12.11 – Part 2

Translation - Termination

- The presence of a stop codon in the A site of the ribosome causes translation to terminate
- The completed protein is released

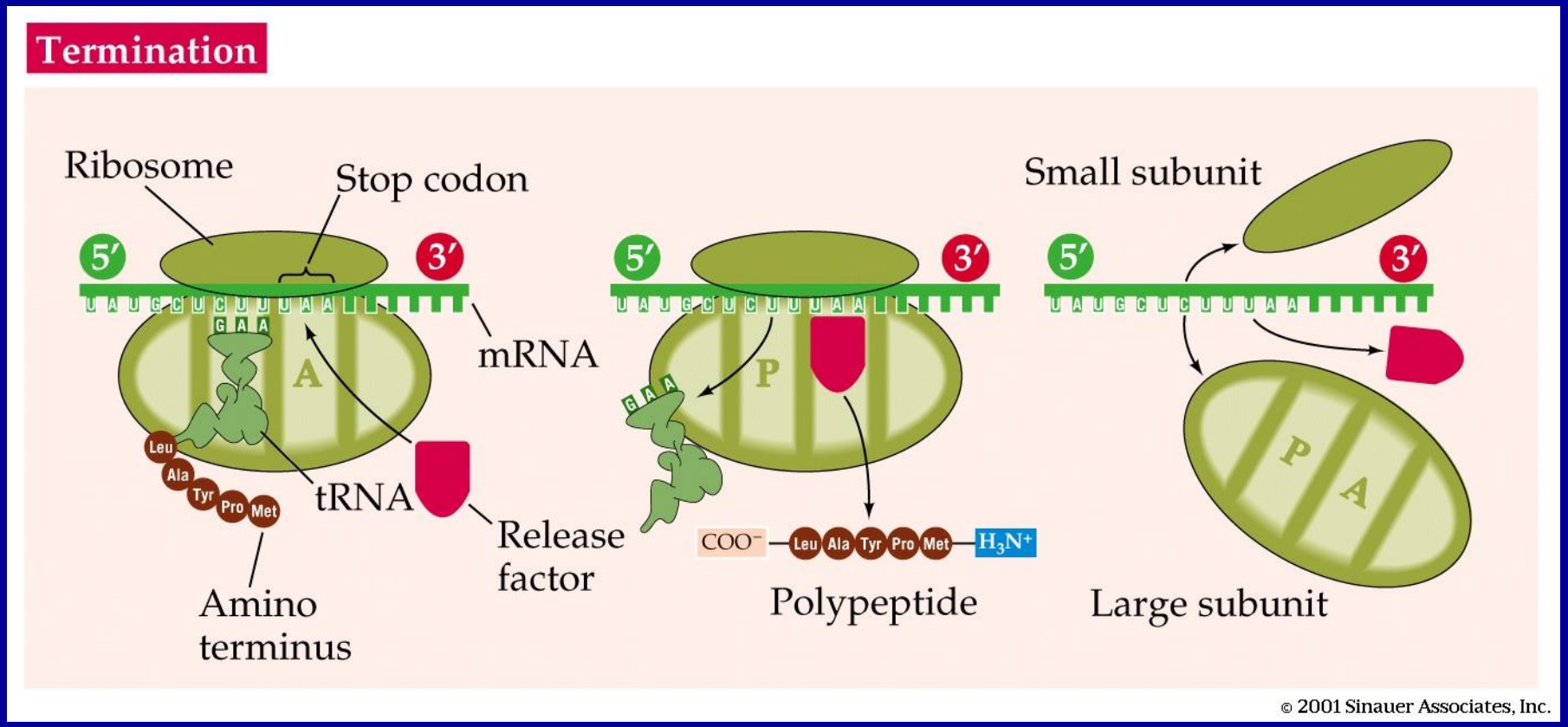
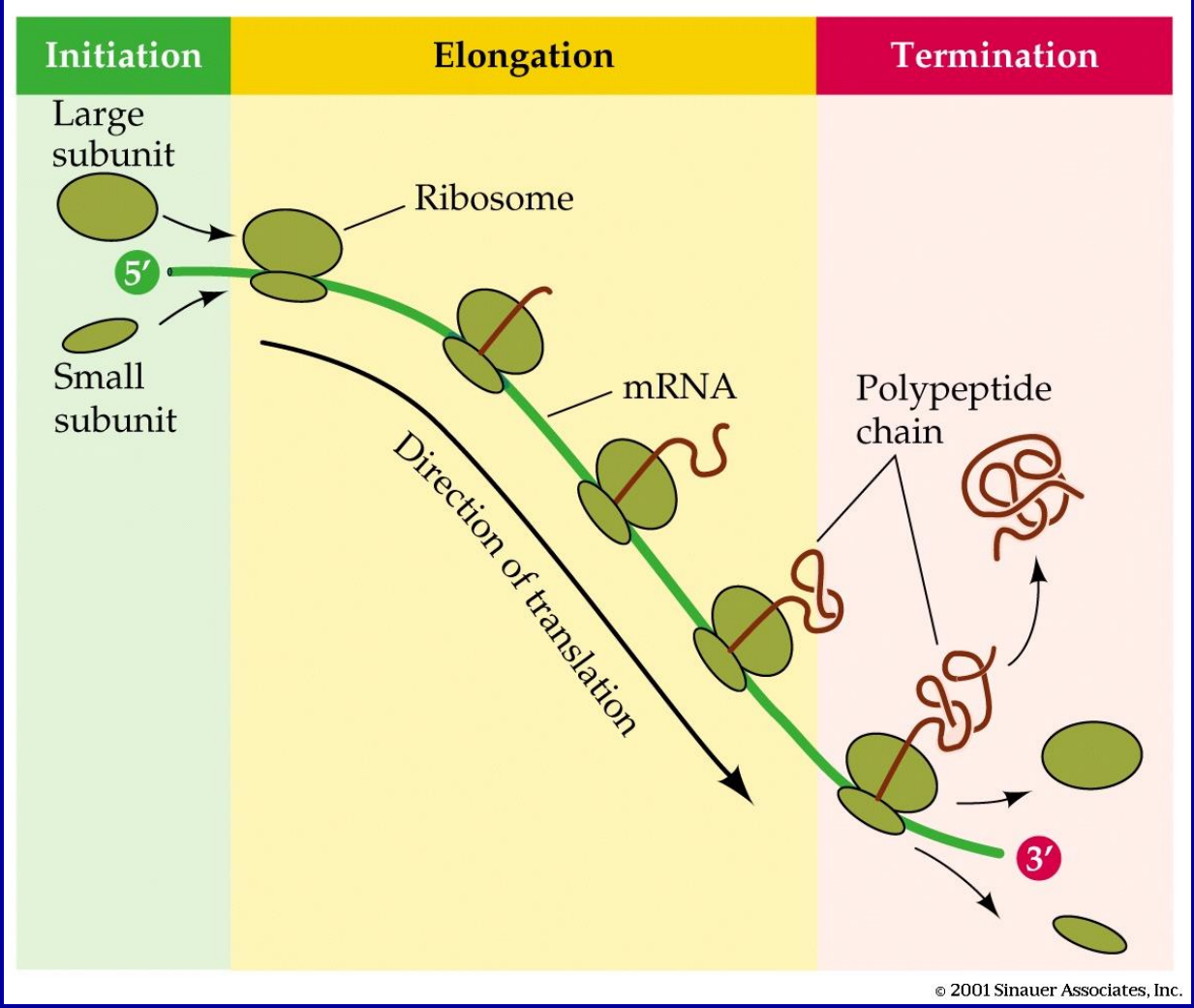


Figure 12.12

Translation



- In a polysome, more than one ribosome moves along the mRNA at one time.

Figure 12.13